TRANSMISSION ENGINEERING STANDARD

نقل الكهرباء
National Grid SA

NATIONAL GRID SA
AMMENDMENT-01
S&SD

TES-P-119.20

Revision No. 0

LV CABLE RACEWAYS AND CABLE INSTALLATION PRACTICES

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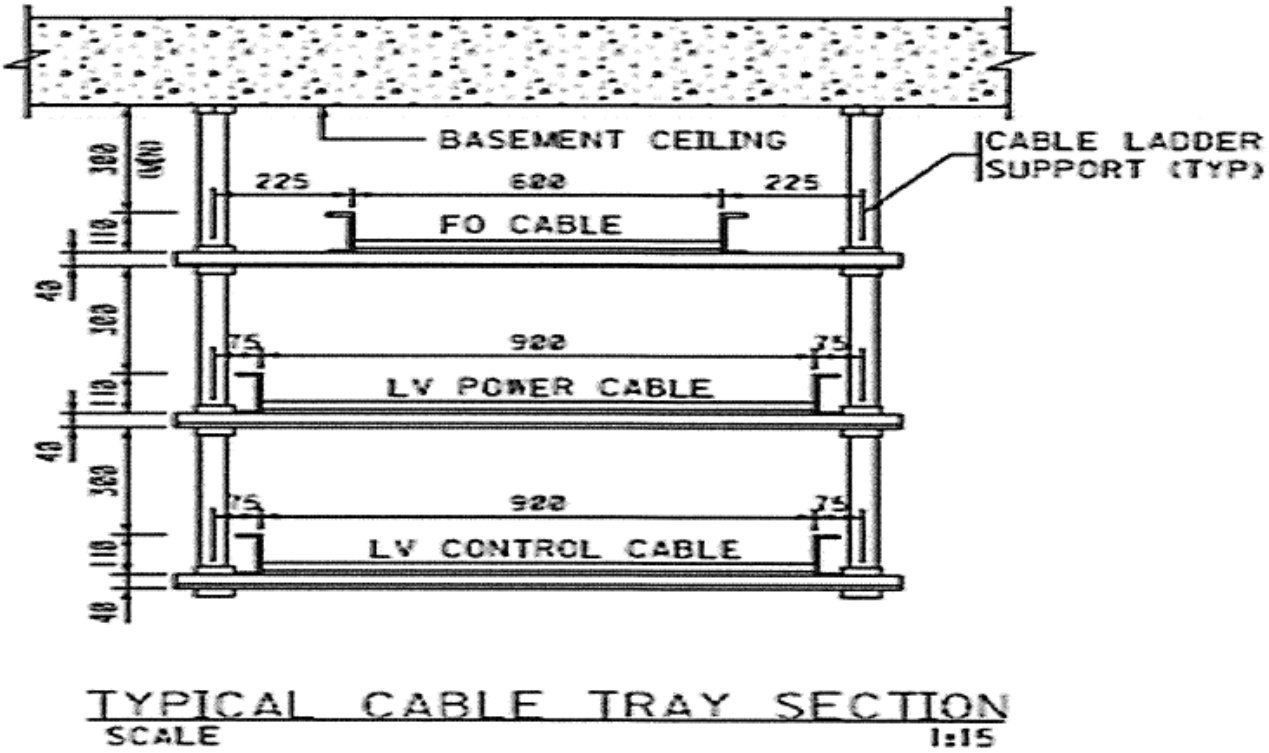
Clause 3.5.4, first paragraph shall be modified as below:

The horizontal and vertical trays shall have **a working** clearance of 300mm between **the top of a tray to the bottom of the upper tray** and tray top to ceiling **as shown in Figures 1 and 2**. **These cable trays can also be arranged in parallel/horizontal depending on the site condition, but still subject to the clearance requirements stipulated by this and related standards.** When the total combined width of the trays exceeds 900 mm, these clearances shall be increased by 150 mm **(Figure 3)** .A passage of 450 mm shall be provided after every four adjacent units of 600mm wide cable tray ,horizontally tired **(Figure 4)**. A clearance of 600 mm shall be maintained from the top of the electrical equipment /panels if trays are located over it.

The Figures below shall be added preferably on Pages 19, 20, and 21


BASEMENT CEILING
CABLE LADDER SUPPORT (TYP)
225
600
225
FO CABLE
75
900
75
LV POWER CABLE
75
900
75
LV CONTROL CABLE
300
110
40
300
110
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300
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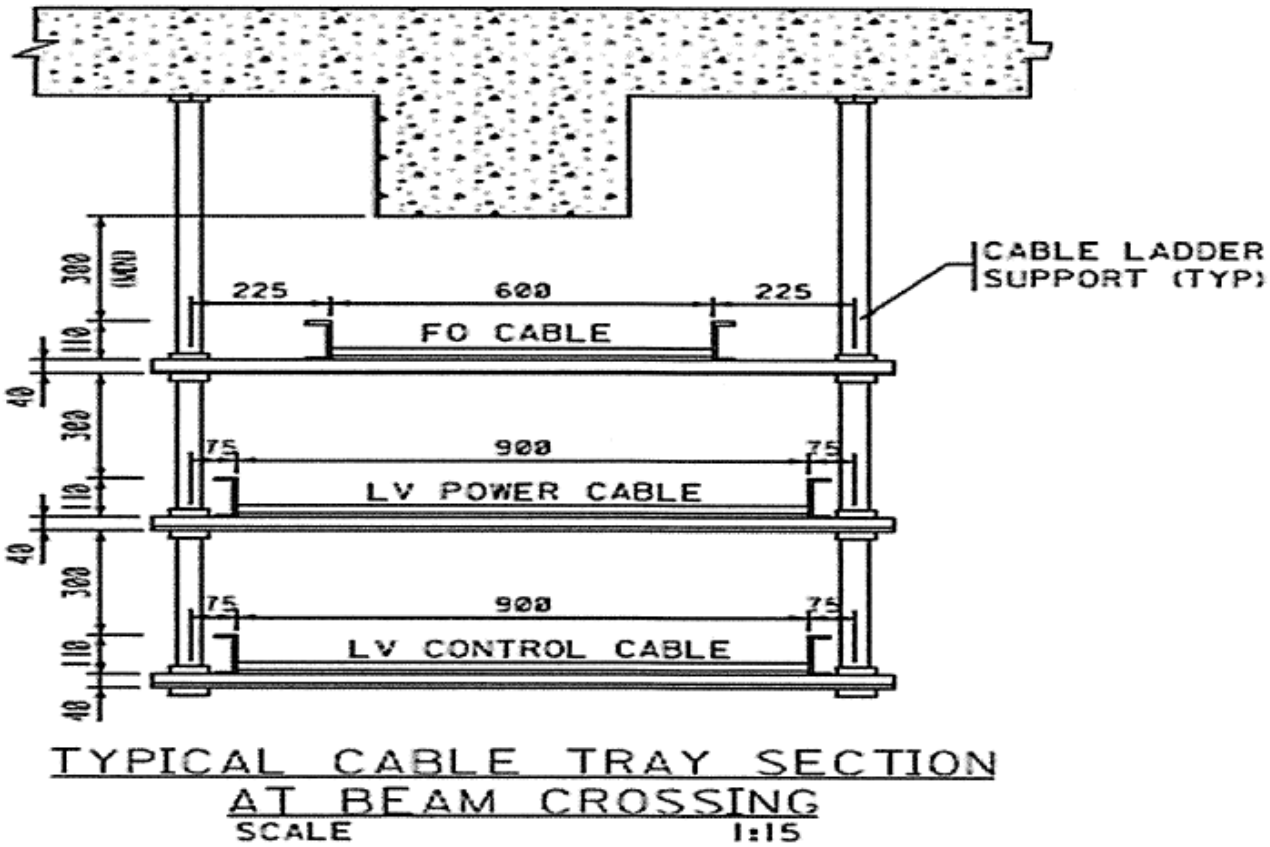
40
TYPICAL CABLE TRAY SECTION
SCALE 1:15

Figure #1


CABLE LADDER SUPPORT (TYP)
225
600
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FO CABLE
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(MIN)
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40
300
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LV POWER CABLE
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LV CONTROL CABLE
110
40
TYPICAL CABLE TRAY SECTION
AT BEAM CROSSING
SCALE 1:15

Figure #2

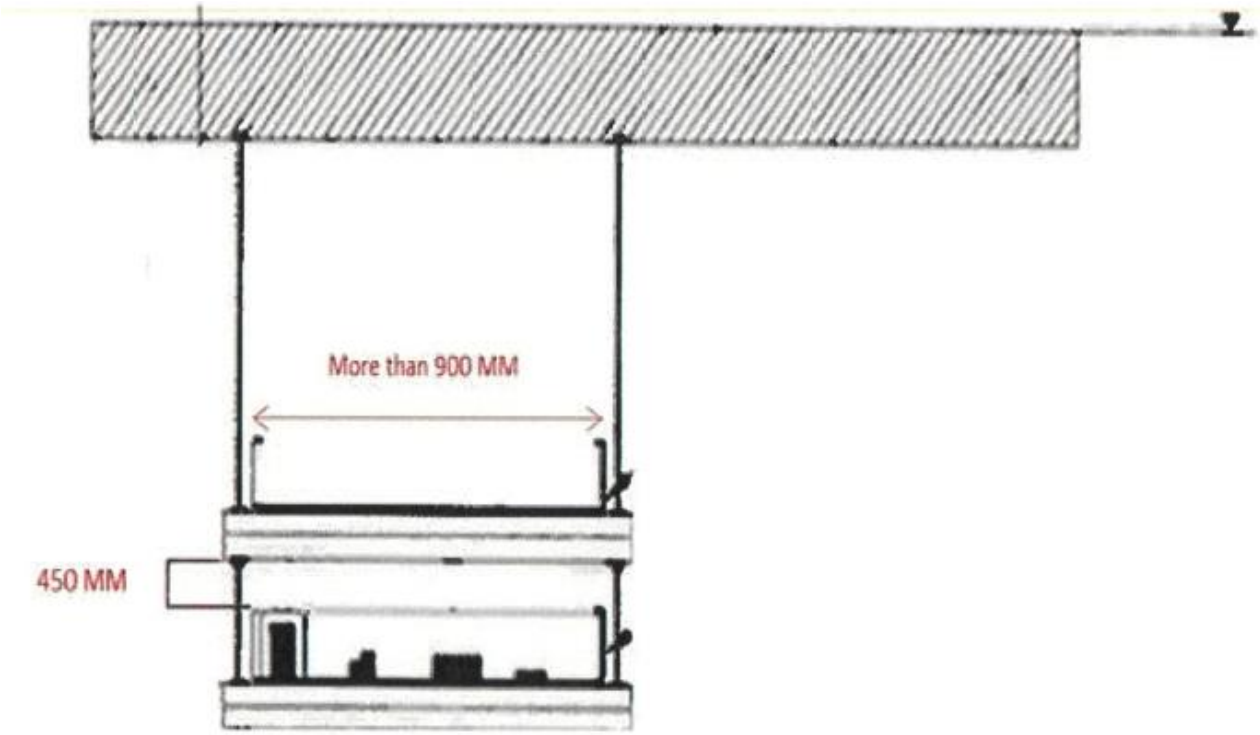

Figure # 3 – When the total combined width of trays exceeds 900 mm, these clearances shall be increased by 150 mm

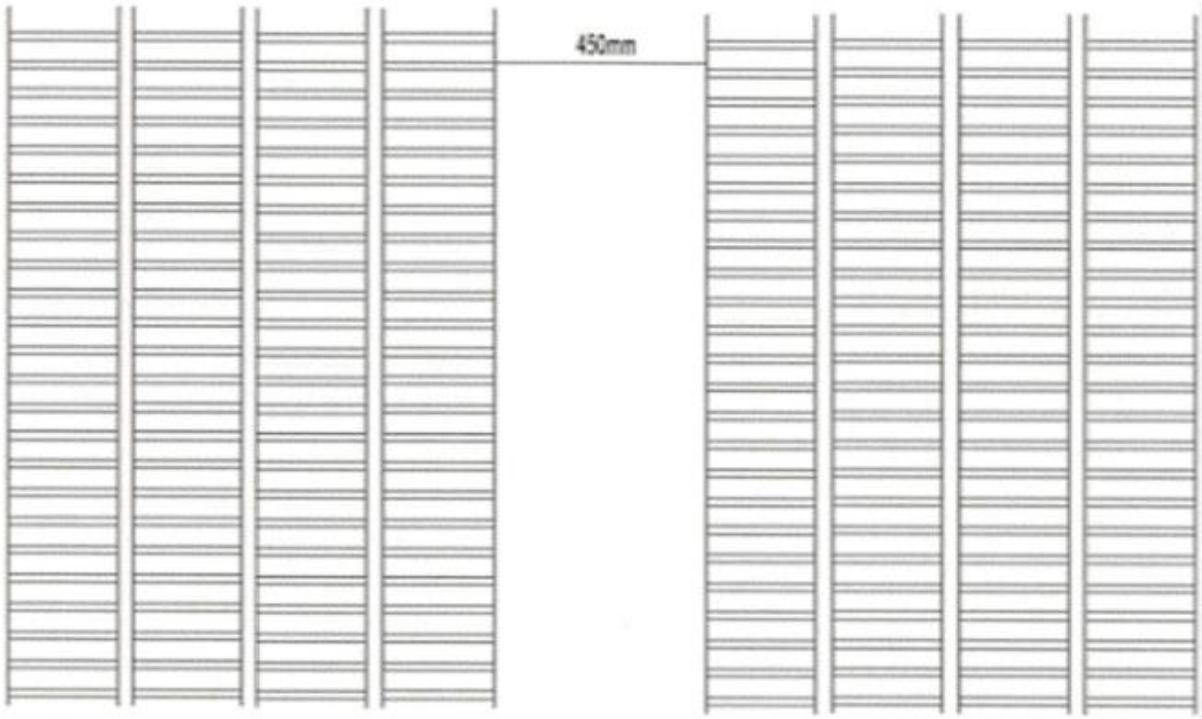

Figure # 4 – A passage of 450 mm shall be provided after every four adjacent units of 600 mm wide cable tray, horizontally tired.

الشركة السعودية للكهرباء
Saudi Electricity Company

REVISIONS				
No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
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Revision no. 0

LV CABLE RACEWAYS AND CABLE INSTALLATION PRACTICES

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

The purpose of this standard is to provide design guidelines for the application of LV Cable Raceways & Cable Installation Practices for LV power and control cables in substations of SEC.

2.0 GENERAL

2.1 Low Voltage power and control cables may be direct buried, installed in duct banks (concrete encased conduits) and trenches below grade, or installed in conduits, trays and raised floor above grade.

2.2 There shall be adequate access for installation of additional cables and removal of cables with ease, economy and minimum wastage.

2.3 The cable system shall be compatible with drainage systems for surface water, oil or other fluids but shall preferably be installed to avoid accumulated fluids.

2.4 Cable entrances to control building shall be provided with non-combustible fire barriers and protection against rodents and water seepage from outside. The cable system shall not propagate fire and shall be consistent with personal safety and good appearance.

2.5 To minimize coupling and promote circuit integrity, separation of low voltage power, control and communication cables shall be done.

2.6 The cable system shall be designed so that foreseeable electrical transients will not adversely affect the cable, connected apparatus or operation.

2.7 Apart from design engineering and material cost, careful consideration shall be given to the installation cost.

2.8 Cable manufacturers recommended maximum pulling tension and sidewall pressure shall not be exceeded and the minimum bending radius shall not be reduced to avoid possible damage to the cable conductor, insulation, shield or jacket.

2.9 Type of cable raceways shall be as per 3.0.

3.0 CABLE RACEWAYS APPLICATION

The cable system provides LV power and control cables for power supply from source to electrical equipments such as transformers, circuit breakers, switching devices, current and potential transformers and auxiliary equipments.

Raceways in the form of conduit, duct bank, trench and tray shall be provided in substations as carrier of cables duly protected and electrically segregated.

3.1 Direct Burial

Direct burial of cables is a method whereby cables are laid in an excavation in earth with cables branching off to various pieces of equipments. The excavation is then backfilled .A layer of sand of minimum width 150mm below and above the cables shall be provided to prevent mechanical damage. Care must be exercised in backfilling to avoid large or sharp rocks, cinders, slag, or other harmful materials.

Direct buried cables, although the least costly installation method, shall generally be avoided except for temporary installations.

3.2 Direct Buried Conduit

Direct buried conduit for LV power and control cables offers the most economical underground system in terms of cost/benefit ratio. This system shall be applied for extending cables for short runs from handhole or manhole to individual equipment terminal box. The short run shall not have more than three 90^0 bends. Direct buried conduits shall not be used under roadways or transformer track.

Direct buried conduit system shall be of DB PVC or PVC coated rigid steel conduit type. Where corrosive environment or excessive alkaline conditions exist, rigid steel conduit shall be protected by a coat of bituminous paint or similar material.

Direct buried conduit including spare shall be installed in trench .A 150 mm layer of clean sand, free of debris, rocks or sharp objects shall be provided around the conduit. Red concrete tile and cable warning tape shall be installed above the sand layer .A minimum spacing of 38 mm between the conduits shall be maintained. Minimum size of the conduit shall be 25mm diameter for the steel conduit and 50 mm dia for the PVC conduit.

PVC conduit shall not be extended above grade .All conduits above grade shall be rigid steel galvanized (RSGC) with suitable concrete encased transition using male or female PVC adapters.

When cables are installed in metallic conduits, all phases of three-phase AC circuits and both legs of single phase AC circuits shall be installed in a same conduit or sleeve.

3.3 Duct Bank

Duct Bank (Concrete Encased Conduit) shall be applicable for all outdoor underground cable installations involving more than one conduit .Duct bank shall consist of two or more types of EB PVC conduits encased in a concrete envelop which extends a minimum 75 mm on the sides and bottom, and 150 mm on the top of the conduit. Minimum size of the conduit shall be of 100mm diameter. Top surface of the concrete envelop shall be painted red. A cable warning tape shall be installed on the top of the duct bank.

Conduits shall be laid on fabricated plastic spacers sized to the conduit outside diameter and desired separation .Plastic spacers for supporting conduits shall be

installed at intervals not exceeding 2.5 m. Intermediate spacers shall be placed on top of the bottom and successive layers of the conduit .The plastic spacers shall provide a minimum spacing of 50 mm between the conduits .The adjacent conduit couplings shall be staggered.

Concrete encasement shall be poured on firm and level ground having a minimum bearing capacity of 140 kPA (3000 psf).Concrete for duct bank shall be as per soil condition as approved by SEC representative and shall meet the requirements of SEC material standard specification 70-TMSS-03 .The joint between the pourings with appreciable time gap shall be made as near vertical as possible and shall be reinforced with dowel bars. All concrete duct banks shall be reinforced with minimum temperature reinforcement in accordance with clause no 7.12.2.1 of ACI- 318 M-05.

All spare conduits shall have non-deteriorating nylon ropes installed inside for future use. Conduits or duct rungs shall be sloped downwards a minimum of 1:400 ,towards manholes or drain points .A test mandrel (6 mm less in diameter than the nominal inside diameter of the duct) and wire brush cleaner (slightly larger than the duct) shall be pulled through each duct.

The testing and cleaning of a duct bank shall be done the day after the concrete has been poured ,to eliminate any concrete which may have seeped into the ducts .A nylon pull rope shall be installed in each duct after cleaning .End bell shall be installed on each end of duct bank. All cable ducts shall be sealed with duct sealing units.

3.4 Trench

Cable trench, cast into concrete floors, shall be provided inside the substations .The trench shall be installed with removable covers, made of chequered aluminum plate flushed with grade. The covers in the rear of the switchboard shall be of fire retardant material and the holes cut in covers to pass cables shall have the edges covered to prevent cable damage.

Fire resistive barriers shall be provided to separate AC and DC power cables from control cables .The cables shall be laid in ladder type trays installed in the trench with clearance of 75mm from the bottom of the trench.

Distance between buried power and communication cables in trench shall be per standard drawing No SB- 036352.

3.5 Tray

Cable tray systems provides high degree of flexibility ,ease in installation and circuit segregation within one tray .In case of underground cable laying where other types of raceways are not possible ,cable tray system shall be considered.

The cable tray and tray support systems shall be in accordance with SEC material Standard Specification 24-TMSS-02.

3.5.1 Design

Cable tray design shall be free from sharp edges /burrs, and based upon the required loading and the maximum spacing between the supports .Loading shall include the static weight of cables and allowance of 20% for the future expansion, and a concentrated load of 890 N at mid span at the center line of the tray or on either side rail. The tray load factor (safety factor) shall be at least 1.5 times based on collapse of tray when supported as a simple beam.

3.5.2 Load Capacity

Tray Load Capacity is defined as the allowable weight of wires, barrier, cables, and fire protection material to be carried by the tray. This value is independent of the dead load of the cable tray system. The quantity of the cables in any tray may be limited to the structural capacity of the tray and its supports.

3.5.3 Size and fill

The tray shall be sized so that the width is same throughout one continuous run .The number of single and multi-conductor cables in cable trays shall not exceed the requirements of Article 392 of NFPA-70 (National Electrical Code).

Not more than 30% to 40% fill for power and control cables and a 40% to 50% fill for instrumentation cables is suggested. This will result in a tray loading in which no cables will be installed above the top of the side rails of the cable tray, except as necessary at intersections and where cables enter or exit the cable tray systems.

The quantity of cables in any tray may be limited by the capacity of the cables at the bottom of the tray in order to withstand the bearing load imposed by cables located adjacent and above. Cable manufacturer's guidelines for load bearing capacity of the cables shall be considered.

3.5.4 Separation

The horizontal and vertical trays shall have clearance of 300mm between any two tiers and tray top to ceiling. When the total combined width of the trays exceeds 900mm ,these clearances shall be increased by 150mm .A passage of 450 mm shall be provided after every four adjacent units of 600mm wide cable tray ,horizontally tired .A clearance of 600 mm shall be maintained from the top of the electrical equipment /panels if trays are located over it.

A minimum of 50 mm separation shall be maintained between tray side rails for adjacent horizontal trays, and 25 mm separation between any vertical support and tray side rail.

3.5.5 Location, Routing and Protection

The cable tray system shall be coordinated with lighting, heating and ventilation systems and fire protection/fire alarm system.

The cable tray shall run parallel and at right angles to building walls and structures, and be routed directly above each panel row and control gear in case of panels having cable top entry provision . Cable trays in the cable basement shall be provided directly below each panel row or switchgear or control gear, supported from the ceiling for easy entry of cable to the panels through floor opening and gland plate. The cable tray racking in the basement shall be supported by freestanding steel structures in the basement. Cable trays shall be located and routed so as not to limit substation equipment accessibility. Cable tray shall not be installed in close proximity to fire hazard equipment or other heat sources that are detrimental to cables.

Cable trays that enter a building shall be sloped down a minimum of 1:100 away from the building .When passing through firewalls, the cables shall be sealed preferably with approved cable transits or fire retardant covering .Special attention shall be given to the installation of fire stops on vertical cable runs. Requirements of TES-P-119.21 shall be met in this regard.

A concrete curb or metal kick plate shall be provided for cable trays passing vertically through floors or platforms .The tray shall be covered on all sides to a distance of 1.8m above floor or platform.

3.5.6 Segregation

The power cable /instrumentation cable segregation shall be provided with physical barrier of at least one hour fire resistance.

A judicious use of flame retardant cable, proper cable separation in tray, proper cable tray separation, fire resistive barrier or shield and application of fire retardant coating to the cables shall be made to ensure the safety of cable system from fire hazards.

3.5.7 Supports and Fittings

Tray sections shall be supported near section ends and at fittings such as tees crosses and elbows per NEMA VE-2 unless otherwise specified.

Horizontal and vertical tray supports shall provide a minimum tray bearing surface of 45 mm and shall have provisions for hold down clamps or fasteners .In addition, vertical tray supports shall provide secure means for fastening cable trays.

Supports shall be located wherever practicable so that connectors between the horizontal straight sections of cable tray runs fall between the support point and the quarter point of the span.Unspliced straight sections shall be used on all simple spans and on the end spans of continuous span arrangements .Tray

supports shall have a maximum of 6 m spacing on horizontal runs and 2.4 m spacing on vertical runs. Cantilever tray sections shall be limited to 900mm.

Vertical trays shall be secured with H-shaped cover clamps (not clips) ,spaced 1.2 m apart on horizontal tray runs and 600 mm apart for on vertical runs with 2 clamps for each bend.

Dropout fittings shall be provided where required to maintain the minimum cable training radius. Where vertical drops exceed 1.5 m outside vertical elbows, drop out fittings shall be installed at the higher elevation.

Where vertical drops exceed 4.5 m inside, vertical elbows shall be installed at the lower elevation. Vertical and horizontal elbows shall have a minimum radius of 300 mm.

3.5.8 Identification

Cable tray sections shall be permanently identified with the tray section number as required by design/construction drawings .Identification shall be stenciled at intervals not to exceed 6 m.

3.5.9 Grounding

Cable tray systems shall be electrically continuous and grounded as per TES-P-119.10.The tray system, with or without a ground conductor attached to and parallel with the tray, shall be effectively bonded to the substation grounding systems.

3.6 Raised Floors

Raised floors provide maximum flexibility for additions or changes .Raised floors construction may be used in control rooms, relay rooms and communications room with prior approval from SEC.

3.7 Manhole

A manhole serves as a point for cable pulling, to change the direction, and as a place to provide contraction and expansion of the conductors. Manholes shall be provided in a duct bank run wherever cable pulling tension or sidewall pressure is expected to exceed the desired limit.
Reinforcing steel in the manhole walls shall not form closed loops around the individual non-metallic conduit entering the manhole and shall be bonded minimum at two points to the substation grounding system. Non–metallic spacers shall be used .Exposed metals in manholes such as conduits and ladders shall be grounded.

Manholes shall be oriented to minimize bends in duct banks .Manholes shall be provided with means for attachment of cable pulling devices to facilitate pulling of cables in a straight line. Provisions shall be made for ladder type trays to facilitate racking of cables along the walls of manhole. Manholes should have a sump, if

necessary, to facilitate the use of a pump. Manholes and manhole openings should be sized so that the cable manufacturer's minimum allowable cable bending radii are not violated.

Manholes shall be constructed in a manner to prevent the entry of flammable liquids.

3.8 Handhole

A handhole is essentially a miniature manhole installed in the main duct bank run approximately 600 mm below grade, with each side measuring about 600 mm.It serves as a point for branching out cables in a direct burial conduit system. Manhole/hand hole cover shall be painted in accordance with 01-TMSS-01 and marking shall be done for the type of cable to branching out e.g. communication/power/instrumentation etc.

3.9 Conduit Size and Fill

The size of conduit required depends upon the size and number of cables to be installed and the length of straightness of the conduit run.

The minimum size of the conduit required for a given size and number of cables can be determined from Tables 1 and 4, Chapter 9 of NFPA – 70.

4.0 CABLE INSTALLATION PRACTICES

This section provides the guidance for storage, handling and installation of LV power and control, cables.

4.1 Storage

Cable reels shall be stored upright on their flanges and handled in such a manner as to prevent deterioration or physical damage to the reel or to the cable. During storage, the ends of the cables shall be sealed against moisture or contamination.

4.2 Precaution

Cable pulling lubricants shall be compatible with cable outer surface, and shall not set up or harden during or after the installation period. The application of cable pulling lubricant shall be as per manufacturer's instructions.
Pulling winch and other necessary equipment shall be of adequate capacity to ensure a steady continuous pull on the cable. Cable shall not be subjected to a reverse bend as it is pulled from the reel.

A tension measuring device shall be used on runs when pulling force calculations indicate that the allowable stresses may be approached. Turning the reel and feeding slack cable to the duct entrance may change a difficult pull to an easy one. Whenever a choice is possible, cables shall be pulled so that the bend or bends are closest to the reel.

Sufficient cable slack shall be left in each manhole and be temporarily supported so that the cable can be trained to its final location on ladder type cable trays along the sides of the manhole.

The use of single-roller or multi-roller cable sheaves of the proper radius shall be ensured when installing cable around corners or obstructions. The rollers shall be properly placed depending on the size and weight of the cable to prevent the cable from sagging and dragging in the cable tray during the pull.

4.3 Protection

Special care must be exercised during welding and soldering operations to prevent damage to cables. If necessary, cables shall be protected by fire-resistant material.

After cable installation has started, trays and trenches shall be periodically cleaned as necessary to prevent accumulation of debris.

A suitable feeder device shall be used to protect and guide the cable from the cable reel into the raceway. The radius of the feeder device shall not be less than the minimum bending radius of the cable. If a feeder device is not used, the cable shall be hand-guided into the raceway.

Bare wire rope shall not be used to pull cables in conduits. A swivel shall be attached between the pulling eye and the pulling rope. Projections and sharp edges on pulling hardware shall be taped or otherwise covered.

Cables shall be pulled only into clean raceways. A test mandrel shall be pulled through all underground ducts prior to cable pulling.

Cables shall be installed in raceway systems that have adequately sized bends, boxes and fittings. Guides for the number of bends between pull points are given in NFPA 70.

The ends of the cables shall be properly sealed during and after installation in wet locations. Aluminum cables shall be resealed after pulling, regardless of locations.

Cable manufacturer's recommendations for cable pulling and minimum training radius shall be followed.

Guidelines of IEEE 576 shall be followed in this regard.

4.4 Vertical Run Support

The weight of a vertical cable shall not be supported by the terminals to which it is connected. Vertically run cables should be supported by holding devices in the tray, in the ends of the conduit, or in boxes inserted at intervals in the conduit system. Cable installed in vertical cables trays should be secured to the cable tray at least every 1.5 m unless otherwise specified.

4.5 Adding Cables

When additional cables are added to cable trays article 392 of NFPA-70 shall be referred for allowable fill and 24-TMSS-02 for allowable load. If it is necessary to work in the vicinity of energized cables while laying additional cables, all precautions per SEC safety procedures and general instruction (for equipment lockout and tagging) and NFPA70B shall be ensured. If any cables are to be touched or moved they shall be de-energized first.

4.6 Cable Pulling

While pulling cables in underground duct (concrete encased conduit), caution must be observed to prevent pulling tensions on cables from exceeding allowable limits.

4.6.1 Maximum Distance of Cable Pulled in Duct Bank

The maximum distance that a cable may be pulled in duct without damaging it, depends on the following factors:

a. Maximum allowable sidewall pressure of the cable construction
b. Tensile strength of conductor or jacket
c. Coefficient of friction between cable jacket and conduit surface
d. Weight of cable
e. Number, location, angle and radius of bends
f. Slope
g. Lubrication
h. Method of pulling cable (pulling eyes, basket weave/grip, etc.)
i. Limits of cable pulling and reel handling equipment

4.6.2 Maximum Cable Pulling Length

Conduit and duct system design shall consider the maximum pulling lengths of cables to be installed. The maximum cable pulling length of a cable is determined by the maximum allowable pulling tension and sidewall pressure as the pulling length will be limited by one of these factors.

4.6.3 Maximum Allowable Pulling Tension

The maximum allowable pulling tension shall be determined from the following formulae unless otherwise indicated by the cable manufacturer:

When pulling one conductor,

$$T_{max} = K.A$$

When pulling together two or three conductors of equal size,

$$T_{max} = 2\ K.A$$

When pulling together more than three conductors of equal size,

T_{max} = 0.6 n.K.A

Where:

T_{max}	=	maximum allowable pulling tension in Newton (N)
K	=	71 N/mm² for annealed copper and hard aluminum, 53 N/mm²for 75% hard aluminum
A	=	cross sectional area of each conductor in mm²
n	=	number of conductors

a. When pulling, using a pulling eye,

T_{max}	=	22.2 kN for a single conductor cable
T_{max}	=	26.7 kN for two or more conductors cable

b. When using a basket grip applied over the outer jacket:

T_{max} = 4.45 kN for non shielded jacketed cables

Tmax shall not exceed tension limit as determined by formula based on basket weave type pulling grip as described in item 4.5.3(c) below.

c. When using a basket weave type pulling grip applied over a lead sheathed cable, the pulling tension shall not exceed 6.67 kN as determined by the formula:

$T_{max} = Km.\pi.t\ (D - t)$

Where:

T_{max}	=	max. Pulling tension (N)
K_m	=	max. Allowable pulling stress (N/mm²)
t	=	thickness of lead sheath (mm)
D	=	outside diameter of lead sheath (mm)

For lead sheathed cables with neoprene jackets, T_{max} = 4.45 kN.

d. Pulling instructions for coaxial and other special cables shall follow the manufacturer's recommendations.

4.6.4 Maximum Allowable Sidewall Pressure

Sidewall pressure is the radial force exerted on the insulation and sheath of a cable at a bend point when the cable is under tension.
The maximum allowable sidewall pressure is 7.25 kN/m of radius for LV power and control cables, subject to modification by the cable manufacturer.

4.6.5 Expected Sidewall Pressure

The sidewall pressure acting upon a cable at any bend may be estimated from the following equations:

$P = T/R$ for one cable.

$P = \frac{(3c-2).T}{3R}$ for three cables in cradle formation where the center cable presses hardest against the duct.

$P = \frac{c.T}{2R}$ for cables in triangular formation where the pressure is divided equally between the two bottom cables.

Where:

P = sidewall pressure on the critical cable(s), (N/m)

T = total pulling tension leaving the bend (N)

R = radius of bend (m)

c = weight correction factor

The cable manufacturer's recommendations shall be followed for all cable configurations not covered by the formulas given above.

4.6.6 Expected Pulling Tension

a. The expected pulling tension of one cable in a straight section of duct may be calculated from the formula (which does not consider slope):

$T = L.m.g.f.c$

Where:

T = total pulling line tension (N)
L = length of conduit runs (m)
m = mass of the cable(s) per unit length (kg/m)
g = acceleration of gravity (9.81 m/s²)
f = coefficient of friction
c = weight correction factor

The coefficient of friction is usually assumed as follows:

Dry cable or ducts	0.5
Well lubricated cable and ducts	0.15 - 0.35

The weight correction factor (c) can be calculated by the following equations:

- Three single cables in cradled configuration:

$$c = 1 + \frac{4}{3}\left(\frac{d}{D-d}\right)^2$$

- Three single cables in triangular configuration:

$$c = \frac{1}{\sqrt{1-\left(\frac{d}{D-d}\right)^2}}$$

- Four single cables in diamond configurations:

$$c = 1 + 2\left(\frac{d}{D-d}\right)^2$$

Where:

D	=	conduit inside diameter
d	=	single conductor cable outside diameter

b. The expected pulling tension of a cable in an inclined section of duct may be calculated from the following formulas:

T_{up} = m.g.L (c.f. Cos α + Sin α)

T_{down} = m.g.L (c.f. Cos α - Sin α)

Where:

α	=	angle of incline from horizontal

c. For conduit runs containing horizontal or vertical bends the expected pulling tension around a bend shall be determined as follows:

$T_{out} = T_{in}\, e^{c.f.\theta}$

Where:

T_{out}	=	tension out of bend (N)
T_{in}	=	tension into the bend (N)
c	=	weight correction factor
f	=	coefficient of friction
θ	=	angle of the change in direction produced by bend (in radians)

This simplified equation, which ignores the weight of cable, is very accurate for majority of the low voltage cable pulling.

d. Pull points or manholes shall be installed whenever calculations show the expected pulling tensions exceed either the maximum allowable pulling tension or sidewall pressure.

e. Weight correction factors for three sing-conductor cables can be arrived from D/d versus correction factor curve (of IEEE 525) given in figure 20-1 in page 19.

4.6.7 Critical Jamming Ratio

When three cables are pulled into a conduit, it is possible for the center cable to be forced between the two outer cables, while being pulled especially around the bend, if D/d ratio is between 2.8 to 3.0.

Up to a ratio of 2.5, the cables are constrained into a triangular configuration. Between 2.8 to 3.0 ratio, jamming of the cables could occur, and the cables might freeze in the duct causing serious cable damage. To allow for tolerances in cable and conduit sizes, D/d ratios between 2.8 and 3.0 shall be avoided.

4.7 Cables in Tray

Each run of cable tray shall be completed before the installation of cables. Supports shall be provided to alleviate stress on cables where they enter a conduit or other enclosure from the raceway system. Protective conduit bushings shall be provided on all conduit entrances and exits from the tray system.

All cables shall be suitably fastened to the cable tray every 1.8 m on horizontal runs and at least on every 1.5 m on vertical runs and at both horizontal and vertical bends.

To prevent imbalance in the parallel conductors due to inductive reactance where circuits are paralleled, single conductor cables shall be fastened in groups consisting of not more than one conductor per phase or neutral. To prevent excessive movements due to fault current magnetic forces, and to minimize inductive heating effects in tray sidewalls and bottom, cables shall be securely bound in circuit groups.

4.8 Electrical Segregation

4.8.1 In general, to minimize interference, cables installed in stacked cable trays shall be arranged in descending order starting from top most tray, as below.

Single core power cables	(ladder type cable tray)
Multicore power cables	(ladder type cable tray)

Multicore control cables	(ladder type cable tray)
Instrumentation cables	(trough type tray with Perforated bottom)

Control cables shall be provided a separation distance of 300 mm from multicore power cables and 600 mm from single core power cables.

4.8.2 The mode of installation of cables in trays shall be as follows:

Multicore control cables	double layer touching
Multicore power cables upto 16mm²	double layer touching
Multicore power cables 25 mm² to 70mm²	single layer touching
Multicore power cables above 70 mm²	single layer 25 mm spaced
Single core power cables 240/300/400 mm²	trefoil touching

Single core power cables for DC and instrumentation cables shall be laid in trough type perforated bottom cable tray.

4.8.3 In special cases, LV power and control cables may be mixed if their respective diameters do not differ greatly and they have compatible operating temperatures and voltage ratings with the prior approval of SEC. When this is done, the power cable ampacity shall be calculated as if all the cables are power cables.

4.8.4 Conductors of signaling, instrumentation, and communication systems shall not occupy the same enclosure, cable tray, conduit or duct with conductors of lighting, power, 127Vac control, or 24V and above DC relay/control systems. Where such conductors are direct buried or in conduit/duct banks, a minimum spacing of 300 mm between the two groups shall be maintained.

4.7.5 Further subdivision of tray systems shall be provided so that the cables associated with duplicate equipment or redundant control devices shall be routed in separate trays to provide isolation, as practically possible.

4.9 Testing

This section provides requirements for testing of power and control cables after installation but before their connection to equipment, and includes cable terminations and connectors.

The purpose of the tests is to verify that major cable insulation damage did not occur during storage and installation. It should be noted that these tests may not detect damage that might eventually lead to cable failure in service, i.e. damage to cable jacket or insulation shield.

Safety precautions shall be performed, as applicable, in conjunction with the cable manufacturer's recommendations.

4.9.1 LV power and control cables shall be insulation resistance tested prior to connecting to equipment and also functionally tested (at equipment operating voltage) as part of the checkout of the equipment system.

4.9.2 The cable insulation resistance tests shall measure the insulation resistance between any possible combinations of conductors in the same cable and between each conductor and station ground, with all other conductors in the same cable grounded.

The test voltage should be a minimum of 500Vdc. The minimum acceptable insulation resistance shall be per IEC 60502-1 for XLPE cables and IEC60227 for PVC cables.

4.10 Termination

The recommendations of the cable and connector manufacturers shall be followed in terminating and connecting low voltage power and control cables.

Where bending of the cable is necessary in training position, care must be exercised that the bending radius is not less than that recommended by the manufacturer.

All terminations and connections shall be protected from contamination or damage by water or other foreign materials.

Cables shall be permanently identified at each end.

4.10.1 Cable End Treatment

Cable ends shall be arranged with neatness for ease in handling.

The cable shall be held in place with clasps or clamps with a sufficient area of contact between the holding device and the cable so that the cable will not be damaged by deformation due to excess bearing pressure.

The cable sheath end shall be dressed with twine or other binder material to prevent the cable from slipping back into the conduit, and for neatness.

4.10.2 Shield

Termination of cable shield requires careful workmanship. To avoid damage to the conductor insulation, extreme care must be exercised when soldering the shield.

4.10.3 Terminals

Whenever practical, control cables shall be terminated on terminal blocks. In any case, all terminations shall be performed in a neat and workmanlike manner.

Conductors of 10 mm² and smaller shall be terminated with tin plated copper, rectangular, round end, crimp type connectors. Conductors larger than 10 mm² shall be terminated with tin plated copper compression terminators or terminal lugs.

Any spare conductors in control cable, not terminated on terminal block, shall be left long enough and dressed.

Exposed conductor surfaces of low voltage power cables at the terminal lugs shall be covered with suitable insulation.

4.11 Reduction of Transients

Transients may be caused by a lighting stroke, a fault, switching operation and ground grid potential difference and may readily be transferred from one conductor to another by means of electrostatic and electromagnetic coupling which results into undesirable induced voltages.

To neutralize the transient induced voltages, shield of shielded control cables, shall be grounded at both ends. However shields of instrumentation and signal cables shall be grounded at a single point only.

If the cable shield is inadequate to carry the fault current, a separate grounding cable shall be run in parallel to the group of conduits carrying shielded cables, which is grounded and bonded to the shield of control cables at each end .This shall be evaluated on case by case basis as per IEEE 525.

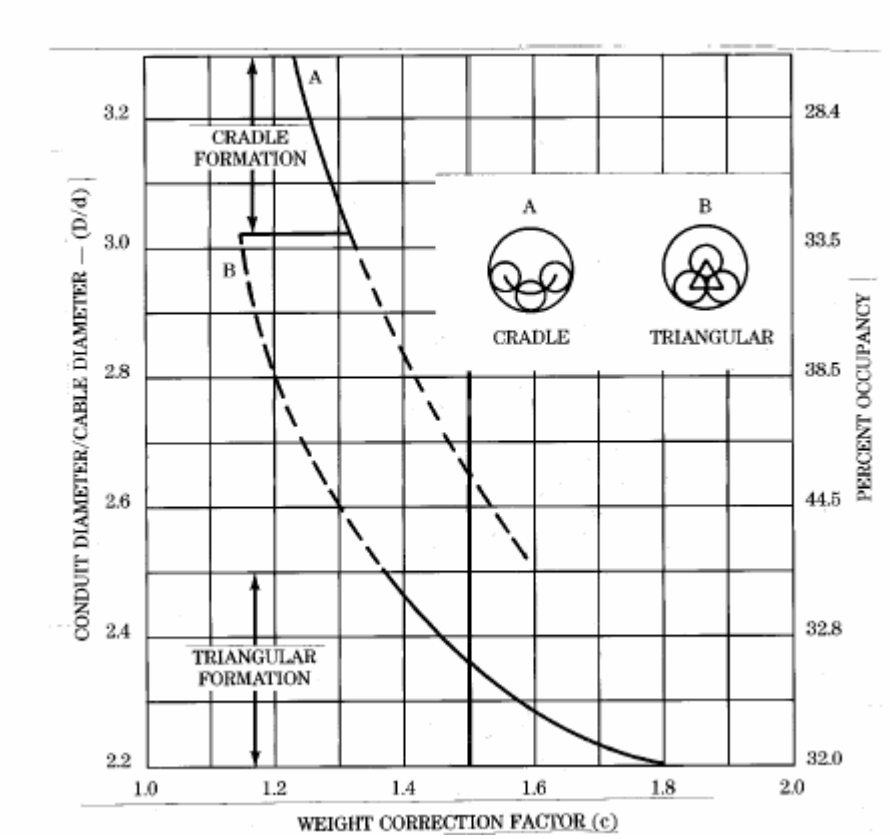

Figure 20-1: Weight Correction factor (c)

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